

Foundation (Jalapeno)

- Q1.** A spacecraft travels 240 miles in 4 hours. What is its unit rate?
(A) 50 miles per hour
(B) 60 miles per hour
(C) 70 miles per hour
(D) 80 miles per hour
(E) 90 miles per hour
- Q2.** A laboratory experiment requires a 3 : 5 ratio of chemical A to chemical B. If 9 units of chemical A are used, how many units of chemical B are needed?
(A) 10 units
(B) 12 units
(C) 15 units
(D) 18 units
(E) 20 units
- Q3.** The force of gravity on an object is directly proportional to its mass. If a 2-kilogram object has a gravitational force of 20 Newtons, what is the gravitational force on a 5-kilogram object?
(A) 25 Newtons
(B) 40 Newtons
(C) 50 Newtons
(D) 60 Newtons
(E) 100 Newtons
- Q4.** A rocket travels 1200 kilometers in 10 hours. What is its average speed?
(A) 80 kilometers per hour
(B) 100 kilometers per hour
(C) 120 kilometers per hour
(D) 150 kilometers per hour
(E) 200 kilometers per hour
- Q5.** The cost of launching a satellite is directly proportional to its weight. If a 100-kilogram satellite costs 1000 to launch, what is the cost of launching a 250-kilogram satellite?
- (A) 1500
(B) 2000
(C) 2500
(D) 3000
(E) 4000
- Q6.** A planet's orbital period is inversely proportional to the square root of its distance from the sun. If a planet with an orbital period of 10 months is 4 astronomical units from the sun, what is the orbital period of a planet 9 astronomical units from the sun?
(A) 12 months
(B) 14 months
(C) 15 months
(D) 17 months
(E) 20 months
- Q7.** A team of astronauts can assemble a space station in 18 days. If 6 astronauts work on the project, how many days will it take for 12 astronauts to assemble the same station?
(A) 6 days
(B) 8 days
(C) 9 days
(D) 10 days
(E) 12 days
- Q8.** A space probe travels 6000 kilometers in 5 hours. What is its unit rate?
(A) 1000 kilometers per hour
(B) 1200 kilometers per hour
(C) 1200 kilometers per hour
(D) 1500 kilometers per hour
(E) 2000 kilometers per hour

Open Ended Questions (FRQ)

- Q9.** A physics experiment involves a direct variation between the force applied to an object and its resulting acceleration. If 5 Newtons of force result in an acceleration of 2 meters per second squared, find the force required to produce an acceleration of 8 meters per second squared. Show all work and explain your answer.
- Q10.** A spacecraft's fuel consumption is inversely proportional to its speed. If 200 kilograms of fuel are consumed when traveling at 5000 kilometers per hour, how much fuel will be consumed when traveling at 10,000 kilometers per hour? Use the given information to write an equation and solve for the unknown fuel consumption. Provide your final answer in kilograms.

Chef's Tips

- Emphasize the importance of units in rate problems.
- Encourage students to draw diagrams to visualize direct and inverse variation.
- Remind students to check their units when converting between different measurement systems.